

Agency MBS Weekly

Taper Time

- **Maintain neutral, but risks higher after FOMC:** We maintain a neutral recommendation on the basis but highlight the increased risks on back of the FOMC meeting earlier this week. There were some negative surprises to our base case, including a higher terminal cap and increased likelihood of a September taper. This will increase the net MBS supply for private market absorption more than we expected, and is a negative for the basis. On the other hand, the surprise on the Treasury caps was even greater, which should be mildly supportive of the basis. Carry in the production coupons also remains quite attractive and MBS offer better relative value than credit, which could be susceptible to a risk-off market reaction.
- **Aggressive taper = more MBS net supply for private investors:** Earlier taper and a higher cap compared to our previous base case will increase the net supply of MBS to the private market even more. If organic net supply remains at \$215bn (our current projection for 2017) in 2017-19, by our calculation the private market will have to absorb \$202bn, \$383bn, and \$449bn, respectively. Investors have not had to absorb >\$300bn supply since 2008 and it is not immediately clear to us which buyers could support these high levels of supply. This remains a key risk to the basis in the coming years.
- **Treasury recommendations for regulatory reform will impact MBS demand:** The Treasury released a highly anticipated report on Monday outlining a list of recommended changes to existing bank regulation. The proposed changes to the SLR (Supplemental Leverage Ratio) and LCR (Liquidity Coverage Ratio) are the most relevant for MBS and Treasury demand. Since these changes do not require Congressional approval, the likelihood of implementation over time is fairly high. Overall, we think that these changes are positive for G-SIB demand for USTs but impact on their MBS demand appears inconclusive to somewhat negative. For non-GSIBs, overall demand for securities is likely to go down but some rotation out of GNs and USTs into FN/FH MBS is possible, which could be a negative for GN/FN swaps.
- **Foreign investor purchases remained muted in April:** The latest TIC data shows that foreign investors bought \$7.7bn agency MBS in April, compared to purchases of \$2.7bn in March. After paydown adjustment, foreign investors net sold \$2.7bn agency MBS, much smaller than the \$8.3bn net selling seen in March. Japanese investors were the larger buyer of MBS in April.
- **Short-term Prepayment Projections:** We expect aggregate Fannie 30yr prepayments to increase by 6% in June, largely driven by a 7% increase in turnover. Day count will remain the same as May (22 days) and refi speeds will likely be marginally higher in June given lower driving rates (-9bp). We expect G2 speeds to increase 7% in June.

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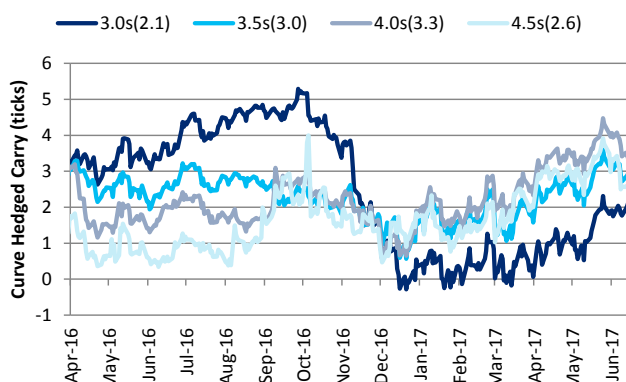
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Market Overview

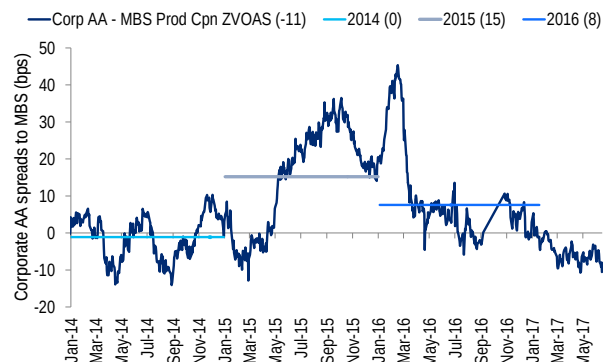
We maintain a neutral recommendation on the basis but highlight the increased risks on back of the FOMC meeting earlier this week. There were some negative surprises to our base case, including a higher terminal cap and increased likelihood of a September taper. This will increase the net MBS supply for private market absorption more than we expected, and is a negative for the basis. On the other hand, the surprise on the Treasury caps was even greater, which should be mildly supportive of the basis. Carry in the production coupons also remains quite attractive (Figure 1) and MBS offer better relative value than credit (Figure 2), which could be susceptible to a risk-off market reaction. In other news, the regulatory reforms proposed by the Treasury should increase the incentive for GSIBs to own Treasuries over MBS, and incentivize non-GSIBs to prefer conventionals over GNs and Treasuries.

Figure 1. Curve hedged carry remains attractive (FNCL)



Source: Citi Research

Figure 2. Corporates spreads are close to the tightest level compared to MBS in recent years.



Source: Citi Research

FOMC puts taper on the table

The FOMC meeting this week yielded some surprises compared to our previous base case expectations for MBS reinvestment tapering, including a higher terminal cap, a lower initial cap, and a longer timeline to reaching the terminal cap. Also, the FOMC statement, Chair's press conference and release of the addendum to policy normalization principles all point to a greater likelihood of a September taper announcement, and Citi economists have revised their base case accordingly ([FOMC – Earlier balance sheet, still looking through soft inflation](#)).

Earlier taper, lower initial cap, higher final cap

The key highlight from the FOMC meeting was the release of an addendum to policy normalization principles, which laid out in some detail the implementation plan for reducing the Fed's balance sheet (Figure 3). While the outline of the Fed's plan is largely similar to what we had initially expected ([Bringing down the Fed's balance sheet](#)), there were some notable differences that we discuss below:

Figure 3. FOMC released implementation guidelines

Summary of Addendum to Policy Normalization Principles and Plans

- * Gradually reduce securities holdings by decreasing reinvestment of principal payments. Paydowns will be reinvested only to the extent they exceed the following caps:
 - Treasuries: Initial Cap of \$6bn/month will increase in steps of \$6bn at three-month intervals over 12 months until it reaches \$30bn/month
 - Agency MBS/debt: Initial Cap of \$4bn/month will increase in steps of \$4bn at three-month intervals over 12 months until it reaches \$20bn/month
 - Caps to remain in place at maximums until Fed is holding no more securities than necessary to implement monetary policy
- * Reduce reserves to a level appreciably below that in recent years but larger than before the financial crisis
- * Fed funds rate is primary means of adjusting monetary policy
 - Reinvestments may resume if material deterioration in economic outlook warrants a sizeable reduction in Fed funds rate
 - FOMC prepared to alter size and composition of balance sheet if economic conditions warrant more easing than Fed funds rate can provide

Source: Citi Research, Federal Reserve

- **The terminal caps are higher than we expected.** The terminal caps of \$20bn for MBS and \$30bn for Treasuries are both higher than the \$15bn cap that we had expected for each. This implies that net MBS supply to the private market will be higher than we had originally expected, at least in the next couple of years. Beyond that, however, the cap on MBS is unlikely to be reached due to declining paydowns on the Fed's portfolio, as we illustrate in the next section. While the same is true for Treasuries based on the current maturity schedule, the Fed will have the flexibility to manage maturities such that the cap is met each month, if they want. This could be achieved by simply reinvesting into shorter-maturity Treasuries going forward.
- **The cap on Treasury portfolio reduction is higher than the MBS portfolio.** We had expected the cap to be similar for MBS and Treasuries, but a higher cap on Treasuries should be a positive for the MBS basis, all else equal.
- **The 12-month timeline to reaching the cap is longer than the 6-months we expected.** We had expected \$5bn increase in MBS and Treasury caps every three months so the announced increase of \$4bn for MBS and \$6bn for Treasuries every three months are marginally positive for MBS versus Treasuries. However, these are only minor deviations from our base case and any benefit is largely offset by higher terminal caps.

On net, the implementation plan details appear to be somewhat negative for the basis, in our view. The increase in MBS net supply for the private market beyond what we expected increases the risk of spread widening although a larger taper in Treasury holdings, which was largely unexpected, provides some offset. More importantly, that the FOMC published the plan at this meeting is bearish for MBS as it suggests that the Fed is very close to beginning the taper. MBS market reaction on the announcement was muted on Wednesday, partially helped by very low origination of just \$1.1bn, but the sector underperformed on Thursday by 4-5 ticks in FN 3.0s-4.0s.

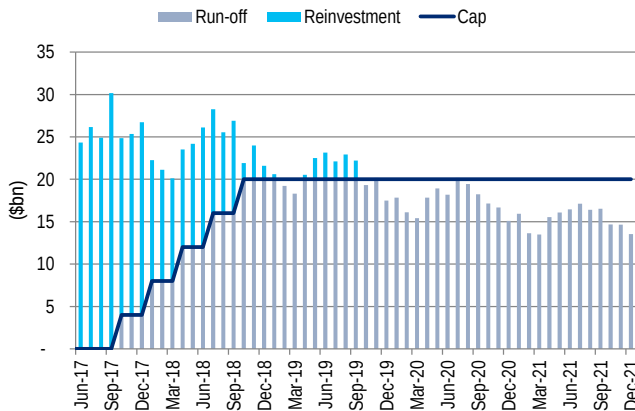
Paydowns to meet the cap amount initially but lag from 2019 onwards

Figure 4 shows our projections for the Fed MBS paydowns and reinvestments at constant mortgage rates based on the newly released plan and Citi economists' forecast for a September taper announcement. As the caps are still rising through 2018 to the terminal level, paydowns are projected to be always higher than the caps.

However, starting in 2019, paydowns undershoot the terminal cap sporadically, and then consistently from 2020 onwards. At constant rates, prepayment speeds will be lower due to burnout and also in winter months due to turnover seasonals. More

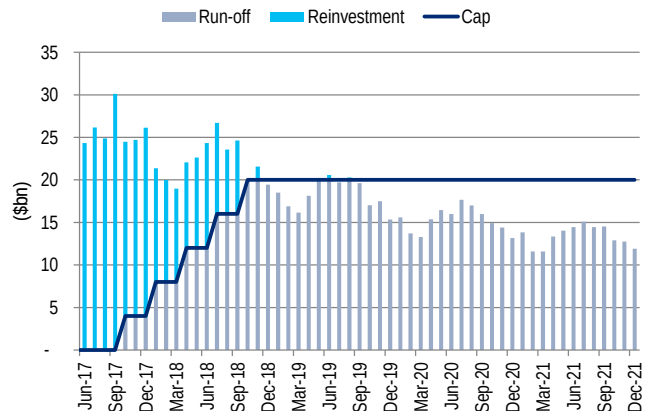
importantly, as the portfolio size declines due to the taper, paydowns will become proportionally lower as well. Paydowns will be even lower if rates rise as implied by forwards, due to a decline in speeds into higher rates (Figure 5).

Figure 4. Fed agency MBS+debt reinvestment and run-off rate based on static yield curve



Source: Citi Research, Yieldbook

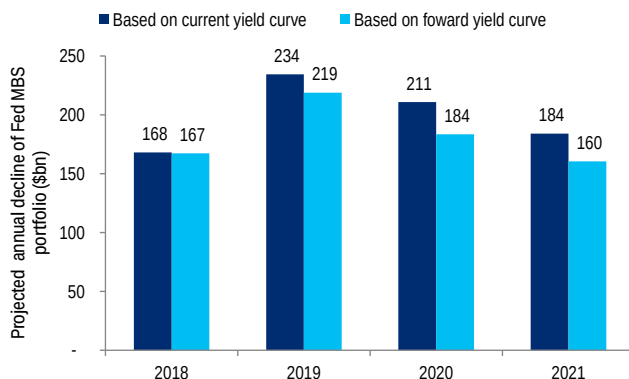
Figure 5. Fed agency MBS+debt reinvestment and run-off rate based on forward yield curve



Source: Citi Research, Yieldbook

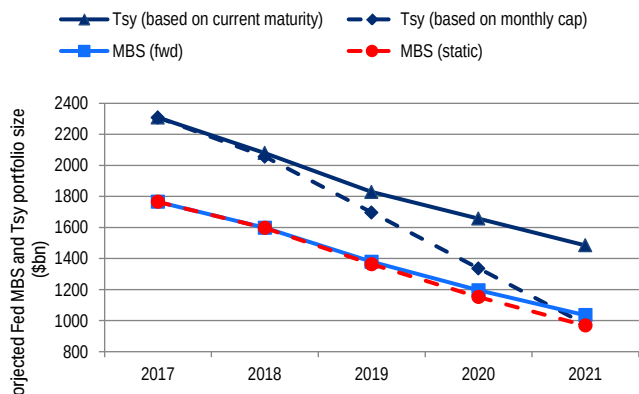
Figure 6 illustrates the annual rundown in the Fed's MBS portfolio through 2021, and Figure 7 compares the cumulative rundown with the Treasury portfolio. While the decline in the Treasury portfolio is comparable to MBS under the current maturity schedule, the risk is to a greater decline in Treasuries as the Fed can choose to reinvest in shorter-dated maturities going forward, thereby increasing the amount of run-off. Note that while these projections are a good representation of our expectations, there will be some deviation in reality as the reinvestments will create collateral differences in future portfolio compared to the current portfolio, which we use for our projections. Any changes in the prepayment and cashout environment will also have an impact on these projections.

Figure 6. Fed MBS portfolio annual run-off projections, assuming a September taper



Source: Citi Research, Yieldbook

Figure 7. Fed MBS (based on static and forward yield curve) and Treasury portfolio sizes (based on current maturity schedule and monthly cap)

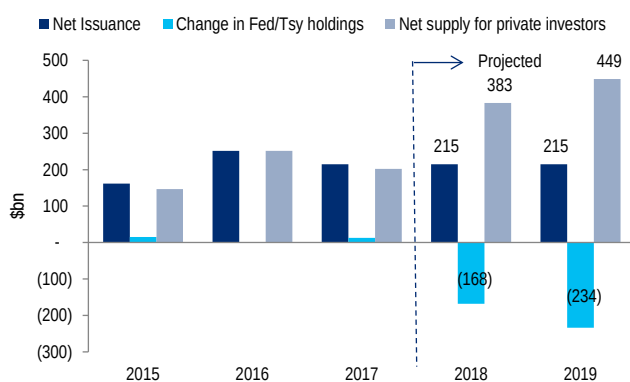


Source: Citi Research, Yieldbook

Private sector MBS net supply to increase by \$12bn, \$168bn, \$234bn in 2017-19, respectively

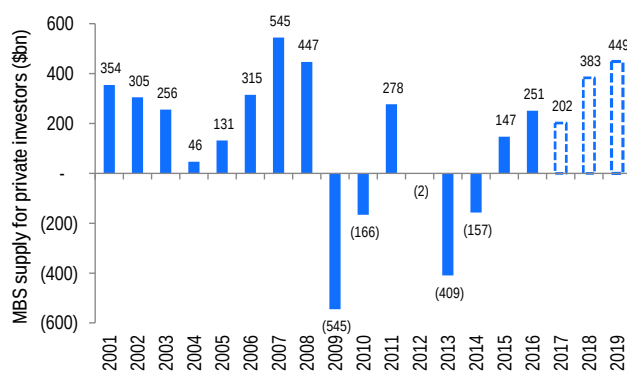
The more aggressive base case taper scenario post-FOMC increases the net supply to the private market that we expect. 2017, 2018, and 2019 net MBS supply for the private market is now expected to increase by \$12bn, \$168bn, and \$234bn, respectively. Compare this with zero, \$135bn, and \$180bn in our previous base case ([Climate Change for MBS](#)). If organic net supply remains at \$215bn (our current projection for 2017) in all of these three years, the private market will have to absorb \$202bn, \$383bn, and \$449bn in 2017-19, respectively (Figure 8). Figure 9 puts these amounts in historical context over the longer term.

Figure 8. Net supply for private investors is expected to increase to post-crisis highs



Source: Citi Research, Yieldbook

Figure 9. Net supply for private investors (ex-Fed/Treasury) since 2001



Source: Citi Research, Yieldbook

Clearly the private market has not had to absorb such high levels of supply in a very long time. Looking at the various buyer contingents, it is not immediately clear to us who could support these high levels of supply. Banks and foreign investors would be the most likely buyers, but their demand remains far from certain. While Japanese investors' demand could be impacted by the recently proposed IRRBB regulation as we discussed last week ([We Need Demand, We Expect Demand](#)), the Treasury regulatory proposals this week also inject uncertainty into US bank demand outlook as we discuss next.

Treasury regulatory proposals and MBS demand

The Treasury released a highly anticipated report on Monday outlining a list of recommended changes to existing bank regulation. Although the report covers a wide array of topics, the proposed changes to the SLR (Supplemental Leverage Ratio) and LCR (Liquidity Coverage Ratio) are the most relevant for MBS and Treasury demand. Since these changes do not require Congressional approval, the likelihood of implementation over time is fairly high. We summarize these changes in Figure 10 and then discuss their impact on MBS demand.

Overall, we think that these changes are positive for G-SIB demand for USTs but impact on MBS demand is inconclusive to somewhat negative. For non-GSIBs, overall demand for securities is likely to go down but some rotation out of GNs and USTs into FN/FH MBS is possible, which could be a negative for GN/FN swaps. It is important to note that we focus only on the impact of these regulatory changes in this article.

Overall bank demand for MBS will be a function of regulations as well as other factors such as deposit and loan growth, incremental spread over Treasuries, etc.

Figure 10. Summary of SLR and LCR changes proposed by the Treasury

Proposed Changes	Regulators
SLR: Significant adjustments should be made to the calculation of the SLR. In particular, deductions from the leverage exposure denominator should be made, including for: (i) cash on deposit with central banks; (ii) U.S. Treasury securities; and (iii) initial margin for centrally cleared derivatives. The eSLR surcharge of 2% should be reevaluated	FRB, FDIC, OCC
LCR/HQLA: The U.S. LCR should be limited to G-SIBs, and a less stringent standard should be applied to internationally active bank holding companies that are not G-SIBs. High-grade municipal bonds should be categorized as Level 2B liquid assets (rather than generally not being counted as HQLA currently). In addition, improvements should be made to the degree of conservatism in cash flow assumptions incorporated into calculations of the LCR to more fully reflect banks' historical experience with calculation methodologies.	FRB, FDIC, OCC

Source: Citi Research

Impact of Supplemental Leverage Ratio (SLR) Changes

The Treasury has proposed that the leverage exposure denominator should exclude cash, U.S. Treasuries and initial margins for centrally cleared derivatives. We estimate that roughly 12-15% of domestic bank assets will be excluded from the SLR calculation due to this change (\$1.4 trillion of cash and \$500bn of Treasuries from total on and off balance sheet items of \$14-17 trillion). This would increase SLRs by 13-17%, or approximately 65-80bp. This improvement in SLR should create room to increase leverage and could therefore generate more demand for securities in general, and Treasuries in particular. All else equal, a more relaxed SLR requirement should also incentivize banks to increase their repo operations.

Almost all U.S. regulations put U.S. Treasuries at par with GNMA MBS, so it is a bit surprising to see that the suggested changes to the SLR calculation exclude Treasuries but not GNs.

G-SIBs: Any relaxation in eSLR would allow GSIBs to increase leverage, which could result in more demand for securities including MBS, all else equal, to increase NIMs. Even if eSLR is left unchanged, GSIBs will be able to increase leverage through buying Treasuries because Treasuries are excluded from the SLR calculation. Furthermore, this exclusion should increase the GSIBs incentive to use Treasuries to meet LCR needs rather than MBS. Overall this should be a clear positive for UST demand, but impact on MBS demand is inconclusive.

Non-GSIBs: Non-GSIBs will be able to increase leverage through owning more Treasuries. Their MBS demand should not be directly impacted due to proposed SLR changes, beyond the ability to increase leverage due to improvement in SLRs from changes to the leverage exposure denominator discussed above.

Impact of Liquidity Coverage Ratio (LCR) Changes

The Treasury recommended that LCR requirements be limited to the G-SIBs, and a less stringent standard be applied only to internationally active non-GSIB bank holding companies.

G-SIBs: LCR will continue to apply. However, if the cash outflow assumptions incorporated into calculations of the LCR are relaxed then need for holding securities for LCR purposes could decline. This is a challenge for MBS demand, on top of the likelihood that G-SIBs will prefer to use Treasuries over MBS for LCR purposes due to favorable SLR treatment as discussed above. Furthermore, inclusion of municipals as new Level 2B HQLA will also create competition for conventional MBS.

Non G-SIBs: LCR will not apply to non-GSIBs that are not internationally active and therefore demand for HQLAs should decline at these banks. Even for internationally active non GSIBs, less stringent standards will pressure demand lower. However, Figure 11 shows how HQLA holdings of non-GSIBs have changed since the LCR regulation was introduced by the Fed in Q3 2013. Treasury and GN MBS, both level 1 assets for LCR, have increased by \$102bn and \$95bn respectively. FN/FH MBS holdings have also increased by \$156bn. If large subsets of these non-GSIBs are exempt from LCR, they may rotate out of Treasuries and GNs and add FN/FH MBS, even if the overall securities portfolio declines due to LCR exemption.

Figure 11. Change in HQLA holdings of non-GSIB US banks from Q3 2013 to Q1 2017

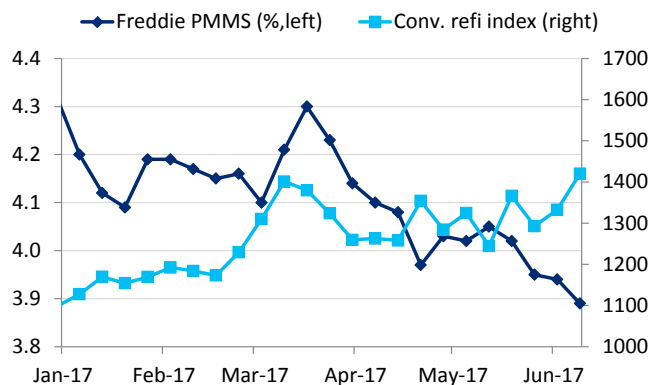
	Cash/Reserves (Level 1)	GNs (Level 1)	Tsy (Level 1)	FN/FH MBS (Level 2A)	Total HQLA
Change	161	95	102	156	490

Source: Citi Research, SNL

Prepayment risks on the horizon

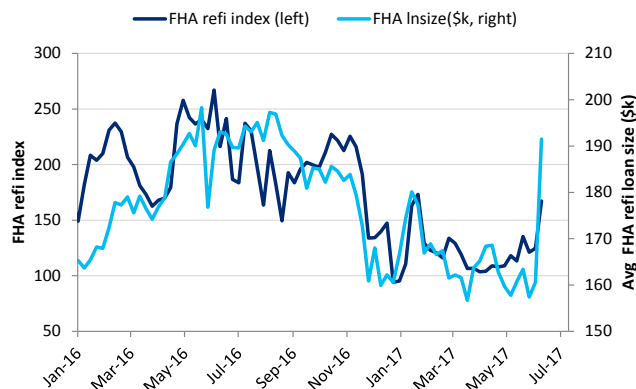
Refinancing risks for MBS are also looming on the horizon as the 10y Treasury yield reached a new year-to-date low this week while the MBA conventional refi index reached a new YTD high (Figure 12). While part of the jump in the MBA index simply reflected a rebound from the previous holiday-shortened week, the index is likely to increase even further next week on back of the rally in rates this week. VA index also saw a small increase.

Figure 12. The conventional refi index reached the highest level YTD as rates continued to rally



Source: Citi Research, MBA

Figure 13. The FHA index jumped this week, along with its average loan size



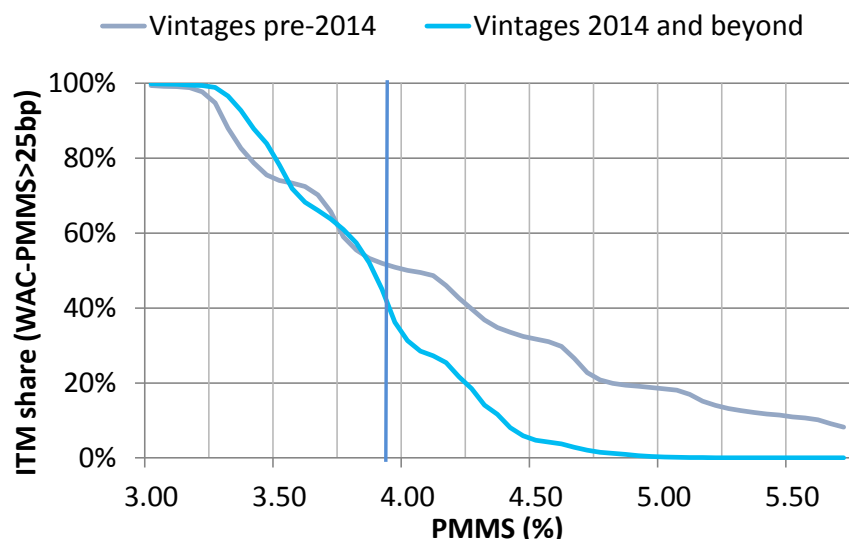
Source: Citi Research, MBA

Most notable, however, was the jump in the FHA index to levels rivaling the January print, when the FHA had announced MIP reductions, which were subsequently rolled back (Figure 13). Loan size also increased sharply from 160K to 192K. It is not clear to us why the FHA index would jump so much more than the conventional and VA indices. Most likely this just reflects a small noisy sample of banks in the index.

Another admittedly far-fetched explanation that may have added to the index would be that originators are not offering HARP as aggressively due to the approaching expiry and no clarity on extension from the FHFA. Therefore, some HARP borrowers may be refinancing into FHA loans. Once again, this is not very likely as there is enough time for servicers to close and deliver HARP loans to GSEs before

expiry. Also, this explanation does not fit with the increase in average loan size. However, given the small sample of banks underlying the FHA index, this kind of substitution could at least theoretically have some impact on the index as we move closer to HARP expiry.

Figure 14. Recent vintages are significantly more cuspy than seasoned vintages at the current mortgage rate.



Source: Citi Research, 1010data

To gauge the potential increase in refinancing activity into a further rate rally, we show the percentage of conventional 30y MBS universe refinanceable at various mortgage rates in Figure 14. We divide the universe into the more recent vintages, which should be more cuspy, versus the less cuspy pre-2014 vintages. Because of greater production in the lower coupons, the curve is more steep for recent vintages, and share of refinanceable borrowers can increase sharply even into a small rally. For instance, the percentage of recent vintages refinanceable increases from 45% at current mortgage rate to 61% into a 25bp rally in mortgage rates.

Summary of views

We present a summary of our views across the agency MBS sectors in Figure 15.

Figure 15. Summary of views across agency MBS sectors

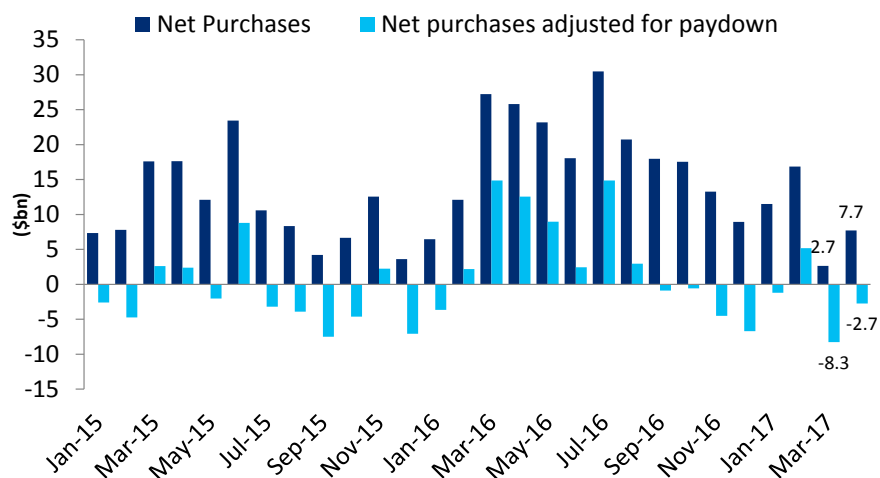
	Summary View/ Risks	Reference Report
Basis	We maintain a neutral stance on the basis but risks have increased since the FOMC meeting	Current report
Trade Recommendations	None	
Supply	Base case net supply forecast revised to \$215bn in 2017. Gross supply of \$1.23 trillion in 2017, 4% mortgage rate.	Agency MBS Weekly: In a New Hawk Minute
Demand	We expect banks to add \$100bn and overseas to add \$30bn in 2017. We upgraded our REIT demand expectations to \$10-20bn. Money managers and insurance companies will need to add \$70bn to balance supply/demand.	Agency MBS Weekly: The New Neutral
Speed Projections	FN 30yr –Jun: +6% July: -9% and G2s –June +7% July: -6%	Current report
Coupon Stack	Prefer FN 3.5s to 3.0s and 4.0s, new issue G2 4.0s still attractive but take profits vs TBA	Agency MBS Weekly: We Need Demand, We Expect Demand
G2/FN	Move from underweight to neutral on G2/FN 3.0s	Agency MBS Weekly: We Need Demand, We Expect Demand
Specified Pools	Neutral on specified pools vs TBA.	Agency MBS Weekly: The New Neutral
Short duration	Neutral on 15s/30s; Underweight ARMs, 10y/20y MBS vs 15s; 10/1s and 7/1s look fair to 5/1s. Quicken pools offer attractive spread pickup but susceptible to rate rally	Agency MBS Weekly: Climate Change for MBS

Source: Citi Research

Foreign investor purchases remained muted in April

The latest TIC data shows that foreign investors bought \$7.7bn agency MBS in April, compared to purchases of \$2.7bn in March (Figure 16). After paydown adjustment, foreign investors net sold \$2.7bn agency MBS, much smaller than the \$8.3bn net selling seen in March.

Figure 16. Net purchase of agency MBS after paydowns was -\$2.7bn from foreign investors, higher than March but remained negative



Source: Citi Research, TIC

Figure 17 and Figure 18 break down agency MBS+debt (without paydown adjustment) and Treasury purchases by country.

- Japanese investors were the larger buyer of MBS in April, purchasing \$6.2bn before paydown adjustment. This is the largest purchase we have seen since November. However, we do not expect this trend to continue and caution against

reading too much into this number. As we noted in our [Agency MBS Weekly](#) on April 7, Japan's foreign investments in late March and early April tend to be volatile due to its fiscal year-end.

In our last [Agency MBS Weekly](#), we also commented on the potential impact on Japan's foreign demand from FSA's new regulation on interest rate risk in the banking book (IRRBB), which goes into effect in FY 2018. In short, the new rule may incentivize Japanese banks to hold more JGBs and shorter duration assets.

- China net sold \$3.4bn MBS+debt and \$3.7bn Treasuries in April. Despite muted demand in April, our outlook for China's demand has improved given its higher foreign exchange reserves and increasing allocation into dollar assets in the last few months. In fact, the more timely Fed foreign custody data suggests that China likely bought close to \$10bn MBS in May and we expect the next release of TIC data to confirm those purchases. See our [Agency MBS Weekly](#) for detailed discussions on our improved outlook for China's demand.

Figure 17. Net purchase of agency MBS+debt breakdown by country (\$bn).

	Japan	China	Taiwan	UK	Others	Total
May-16	9.2	2.8	2.3	7.9	2.4	26.8
Jun-16	9.6	1.2	3.4	2.8	16.5	24.9
Jul-16	13.9	6.4	3.2	2.3	12.6	33.4
Aug-16	11.6	0.1	3.4	3.8	10.0	24.0
Sep-16	9.4	2.1	4.8	2.9	12.9	23.9
Oct-16	9.5	2.9	4.7	1.8	7.4	22.1
Nov-16	4.8	3.4	3.3	(0.5)	4.5	15.7
Dec-16	0.7	(0.2)	3.3	0.6	9.3	12.7
Jan-17	0.7	0.1	6.0	0.6	6.8	12.6
Feb-17	(0.5)	12.9	4.0	1.6	(4.9)	12.7
Mar-17	1.3	(2.2)	4.5	(0.8)	(3.9)	(1.1)
Apr-17	6.2	(3.4)	3.1	0.5	(2.1)	4.2
YTD 2017	7.7	7.4	17.6	1.9	-4.1	28.4

Note: Including both agency MBS and debt but paydowns are not included
Source: Citi Research, TIC

Figure 18. Net purchase of Treasuries breakdown by country (\$bn)

	Japan	China	Taiwan	UK	Others	Total
May-16	5.8	4.4	(3.1)	26.4	(53.6)	(19.9)
Jun-16	(13.2)	(28.0)	0.5	5.7	7.7	(26.8)
Jul-16	3.6	(21.2)	2.4	(0.3)	1.9	(12.8)
Aug-16	(9.3)	(19.2)	(1.4)	(8.4)	13.7	(22.8)
Sep-16	(18.7)	(14.9)	(1.3)	(22.2)	(19.1)	(74.3)
Oct-16	(16.1)	(25.7)	1.0	12.9	(31.8)	(59.8)
Nov-16	(6.7)	(2.9)	1.9	11.7	(3.7)	0.3
Dec-16	(14.4)	(1.3)	1.3	4.0	(11.6)	(22.5)
Jan-17	(4.4)	1.3	(3.8)	(2.4)	2.1	(7.2)
Feb-17	(1.9)	1.6	(1.1)	11.9	(23.9)	(13.5)
Mar-17	31.2	(3.7)	(0.4)	1.6	(4.3)	24.4
Apr-17	(8.4)	(3.7)	(0.3)	(5.9)	(4.2)	(22.5)
YTD 2017	16.5	-4.6	-5.7	5.2	-30.3	-18.8

Source: Citi Research, TIC

Short-term Prepayment Projections

Figure 19. Aggregate speed projections for FNCL and G2SF

	May CPR	Jun/May	Jul/Jun
FNCL	11.7	6%	-9%
G2SF	14.3	7%	-6%

Source: Citi Research

We expect aggregate Fannie 30yr prepayments to increase by 6% in June, largely driven by a 7% increase in turnover (Figure 20). Day count will remain the same as May (22 days) and refi speeds will likely be marginally higher in June given nearly unchanged conventional refi index but lower driving rates (-9bp). We expect G2 speeds to increase 7% in June.

Fannie 30yr speeds in July are expected to decline by 9% as daycount and turnover should drop by 9% and 6% respectively, more than offsetting a 9% increase of the conventional refi index + loan size. G2SF are expected to decline less (-6%) in July given the recent increase of the refi index.

Figure 20. Driving factors for prepayments (6 week lag)

	Apr	Change	May	Change	Jun	Change	Jul	Change
Day Count	19.5*	-15%	22	13%	22	0%	20	-9%
Turnover seasonal	1.04	16%	1.16	12%	1.24	7%	1.17	-6%
PMMS rate	4.20	0.03	4.11	-0.09	4.02	-0.09	3.92	-0.10
Conventional Refi Index	1,335	14%	1,305	-2%	1,309	0%	1,371	5%
Conventional Insize (\$k)	258	3%	265	3%	268	1%	280	4%
FHA Refi Index	122	-6%	106	-13%	118	11%	147	25%
FHA Insize(\$k)	160	-5%	165	3%	161	-3%	177	10%
VA Refi Index	105	-11%	99	-6%	102	3%	115	12%
VA Insize (\$k)	215	4%	220	2%	223	1%	235	5%

Note: We count good Friday as a half day.

Source: Citi Research, MBA, Freddie Mac

Similar to last month, lower coupons are expected to increase the most across the stack. FN 3.0s are expected to increase by 7% in aggregate. The 2016 vintage should speed up the most (+13%) as they continue to ramp up the seasoning curve. FN 3.5s in aggregate should speed up by 7% with the 2016 vintage up 10% and more seasoned vintages up 5%. FN 4.0s are expected to increase by 5%. High coupons (4.5s and above) may play some catch up in June as their speeds surprised to the lower side last month. However, seasoned high coupons are not expected to completely rebound from the downside surprise in May and we only expect them to increase 4-5%.

In G2s, aggregate speeds in June are expected to increase by 7%, slightly more than conventionals. While the conventional refi index stayed relatively flat for June, the FHA/VA index (index + loan size) moved 4-7% higher after multiple months' of relative weakness, suggesting refi speeds could increase more. The higher refi speeds should occur across G2 3.0s-4.0s for which we expect a 7-9% MoM increase compared to a 5-7% increase in conventionals for the same coupons

Figure 21. Aggregate coupon level projections for FNCL and G2SF

FNCL				G2SF			
Coupon	May CPR	Jun/May	Jul/Jun	Coupon	May CPR	Jun/May	Jul/Jun
3.0	7.8	7%	-8%	3.0	12.3	9%	-7%
3.5	11.3	7%	-7%	3.5	15.7	8%	-5%
4.0	14.9	5%	-9%	4.0	20.9	7%	-5%
4.5	17.3	4%	-10%	4.5	21.3	4%	-7%
5.0	19.2	4%	-10%	5.0	22.3	2%	-8%
5.5	18.9	4%	-10%				
6.0	21.2	4%	-10%				

Source: Citi Research

Figure 22. Short-term prepayment projections for selected Fannie Mae 30-Year vintages, June 17- July 17.

Coupon	Year	Bal (\$bn)	Factor	WALA	WAC	WACLS	FICO	LTV	Actual		Projection	
									Apr	May	Jun	Jul
2.5	2013	3.8	0.74	52	3.15	276	774	67	5.9	8.0	8.4	7.4
2.5	2012	5.1	0.71	55	3.20	275	776	68	6.0	7.7	8.1	7.1
3.0	2016	228.6	0.95	10	3.66	304	760	75	4.1	5.1	5.8	5.4
3.0	2015	52.0	0.80	25	3.77	297	766	74	6.8	9.1	9.7	9.1
3.0	2013	139.7	0.68	50	3.58	265	764	72	8.1	10.3	10.7	9.6
3.0	2012	101.9	0.63	56	3.58	267	770	71	7.7	9.4	9.9	8.9
3.5	2016	109.9	0.91	12	4.08	265	738	78	6.5	8.2	9.1	8.8
3.5	2015	161.0	0.76	23	4.11	271	750	78	9.1	11.4	12.3	11.6
3.5	2014	49.7	0.59	33	4.24	262	759	78	10.9	13.4	14.0	12.9
3.5	2013	69.5	0.60	48	4.02	233	743	77	10.3	12.8	13.4	12.1
3.5	2012	116.1	0.50	60	4.00	233	759	74	10.1	12.0	12.6	11.3
3.5	2011	18.8	0.37	68	4.01	245	772	70	9.4	11.0	11.6	10.4
3.5	2010	7.4	0.32	79	4.10	231	773	69	9.6	12.3	13.0	11.7
4.0	2016	38.5	0.90	12	4.51	236	712	79	8.9	12.2	13.4	12.8
4.0	2015	47.2	0.74	23	4.58	224	716	80	13.2	15.2	16.2	14.8
4.0	2014	77.9	0.53	35	4.58	231	737	80	14.3	16.8	17.5	15.8
4.0	2013	47.4	0.48	45	4.57	216	740	80	13.2	15.9	16.5	14.9
4.0	2012	33.7	0.45	61	4.47	191	737	78	12.3	14.7	15.2	13.7
4.0	2011	36.8	0.31	71	4.46	214	759	73	12.0	13.9	14.4	13.0
4.0	2010	31.1	0.25	80	4.48	213	764	71	11.4	13.3	13.8	12.4
4.0	2009	18.1	0.15	96	4.54	207	765	66	12.9	14.1	14.7	13.2
4.5	2014	14.6	0.54	37	5.02	177	704	81	17.7	20.2	21.3	19.4
4.5	2013	9.6	0.48	44	5.04	176	717	80	16.2	18.1	18.8	17.1
4.5	2011	28.8	0.28	72	4.93	190	746	75	14.7	16.1	16.7	15.1
4.5	2010	26.9	0.20	84	4.94	202	752	74	14.5	16.9	17.5	15.8
4.5	2009	34.8	0.12	95	4.93	196	756	70	15.1	17.1	17.8	16.1
5.0	2011	8.6	0.29	73	5.38	182	726	80	17.8	18.3	19.1	17.3
5.0	2010	14.0	0.21	85	5.37	194	730	79	18.3	20.5	21.2	19.2
5.0	2009	10.6	0.13	94	5.42	179	737	75	18.0	20.3	21.1	19.1
5.0	2008	3.6	0.04	109	5.65	179	733	71	17.9	21.8	22.5	20.4
5.0	2005	7.2	0.05	143	5.64	149	722	70	19.1	19.0	19.8	17.9
5.0	2004	4.5	0.05	157	5.54	131	721	70	16.1	18.0	18.8	16.9
5.0	2003	11.1	0.05	166	5.50	114	719	70	15.9	17.2	17.9	16.1
5.5	2008	5.7	0.04	109	6.04	165	725	74	20.0	20.8	21.6	19.6
5.5	2007	5.3	0.03	120	6.13	166	717	72	21.9	21.3	22.1	20.1
5.5	2006	2.6	0.03	132	6.14	156	715	70	19.8	20.2	20.9	19.0
5.5	2005	7.6	0.06	143	5.98	129	709	73	17.6	18.7	19.5	17.6
5.5	2004	6.6	0.05	155	5.93	119	710	73	16.6	17.7	18.5	16.7
5.5	2003	11.7	0.04	168	5.94	104	712	72	15.2	17.2	17.9	16.1
6.0	2008	3.3	0.04	108	6.54	148	711	77	22.5	22.2	23.1	20.8
6.0	2007	7.2	0.04	119	6.57	147	705	75	21.8	22.8	23.7	21.4
6.0	2006	5.9	0.04	132	6.56	138	705	73	20.6	21.5	22.2	20.2
6.0	2004	3.0	0.05	156	6.44	102	697	78	16.5	18.7	19.5	17.6
6.0	2003	2.6	0.03	168	6.49	95	700	75	14.9	17.5	18.3	16.5

Source: Citi Research

Figure 23. Short-term prepayment projections for selected Ginnie Mae II 30-Year vintages, June 17- July 17.

Coupon	Year	Bal (\$bn)	Factor	WALA	WAC	WACLS	FICO	LTV	FHA%	VA%	Actual		Projection	
											Apr	May	Jun	Jul
3.0	2016	199.6	0.93	10	3.40	275	706	94	47	49	5.6	7.9	9.0	8.7
3.0	2015	64.7	0.76	24	3.46	259	715	95	45	50	10.6	14.1	15.1	14.2
3.0	2013	80.4	0.58	50	3.33	226	718	94	51	42	12.5	15.2	16.2	14.8
3.0	2012	34.8	0.53	56	3.39	218	721	95	54	37	12.6	14.9	15.9	14.4
3.5	2016	130.2	0.87	12	3.88	248	678	94	71	26	9.4	12.0	13.1	13.3
3.5	2015	137.4	0.68	23	3.87	238	692	94	67	29	13.3	16.0	17.4	16.4
3.5	2014	54.1	0.52	34	3.90	218	706	96	41	48	16.4	18.8	20.1	18.7
3.5	2013	41.1	0.50	47	3.85	198	694	94	63	30	14.9	17.2	18.4	16.7
3.5	2012	69.4	0.43	59	3.81	207	711	94	62	32	14.9	17.9	18.8	17.0
3.5	2011	10.7	0.38	67	3.88	195	717	96	54	36	15.2	16.4	17.3	15.8
4.0	2016	17.2	0.81	13	4.39	231	654	95	81	17	21.2	25.5	26.5	27.9
4.0	2015	32.5	0.60	22	4.37	215	664	94	83	13	20.7	21.4	22.8	22.0
4.0	2014	46.8	0.41	35	4.35	185	673	95	66	22	19.6	21.1	22.8	21.1
4.0	2013	20.1	0.37	44	4.36	180	692	95	59	26	19.9	21.3	22.8	21.0
4.0	2012	8.9	0.30	62	4.31	179	676	94	79	17	16.8	17.6	18.9	17.5
4.0	2011	21.6	0.28	71	4.35	175	703	95	66	23	17.3	18.8	20.1	18.6
4.0	2010	9.6	0.28	80	4.38	189	714	95	77	17	15.3	17.9	18.9	17.5
4.5	2014	4.1	0.37	38	4.82	160	648	94	85	11	23.5	25.3	26.6	25.0
4.5	2013	9.2	0.35	44	4.82	157	667	94	76	12	21.6	23.3	25.2	23.4
4.5	2011	15.9	0.22	73	4.81	162	693	95	76	14	18.2	19.8	20.4	18.8
4.5	2010	18.4	0.22	84	4.87	175	702	94	83	11	17.8	20.4	21.1	19.5
4.5	2009	9.7	0.18	95	4.91	164	695	94	81	12	18.7	20.6	21.4	19.9
5.0	2010	13.5	0.18	86	5.29	155	682	94	88	5	19.5	21.4	22.1	20.2
5.0	2009	9.8	0.15	93	5.36	150	676	94	84	6	19.6	23.4	23.6	21.7
5.5	2009	1.6	0.13	94	5.86	126	637	93	90	6	19.9	20.1	19.7	17.9
5.5	2008	2.4	0.06	107	5.95	130	654	93	83	9	23.1	23.7	24.9	22.8
6.0	2008	1.9	0.05	107	6.46	116	644	94	82	6	21.4	22.2	22.3	20.5

Source: Citi Research

Appendix A-1

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